NON-WSQ COURSE |
EMBEDDED SYSTEMS

Arduino Masterclass

Course Code C218

Non-WSQ commercial short course
Conducted by Tertiary Infotech Academy Pte Ltd | UEN 201200696W

SENSE
DECIDE
ACT

Version 1.0 | 30 September 2026

Trainer: _____

About the Trainer - Template

Name

_____

Qualifications

_____

Industry practice

_____

Contact

_____

NAMED TRAINER PROFILE

About the Trainer - Dr Alfred Ang

Role

Principal Trainer, Tertiary Infotech Academy Pte Ltd

Engineering lens

Embedded systems, automation, software and applied AI

Teaching approach

Mechanism first: build, measure, diagnose, then improve

Learner evidence

Working circuits, readable sketches, serial traces and acceptance checks

CONTEXT MAP

Let's Know Each Other

Your role

Where do electronics or automation show up in your work?

Prior exposure

Arduino, C/C++, breadboards, sensors, motors - any level is welcome.

Target build

What real device would you like to sense or control?

Evidence habit

How will you prove a prototype works rather than merely looks complete?

WORKING AGREEMENT

Ground Rules

Safety first

Power OFF before rewiring. Never drive a motor directly from an I/O pin.

One circuit at a time

Build, inspect, power, measure, and record evidence before extending.

Active participation

Predict the output, explain the mechanism, then run the test.

Professional conduct

Respect others, return on time, and keep phones silent.

Course Facts and Time Budget

Course

Arduino Masterclass

Duration

One day: 7.5 instructional hours (9:30am - 5:30pm)

Progress checks

Lab acceptance tests after every topic - no formal exam

Hardware

Arduino Uno/Maker UNO, breadboard, sensors, driver stages and actuators

Course Learning Flow

1

Topic 1

Board, sketch, timing, serial

2

Topic 2

Voltage, current, digital I/O, PWM

3

Topic 3

ADC, calibration, LDR, servo

4

Topic 4

Ultrasonic, PIR, timeouts

5

Topic 5

DHT, drivers, motors, feedback

SAFETY CONTROL POINTS

Equipment and Electrical Safety

I/O boundary

Use logic pins for signals, not motor/servo supply current.

Power boundary

External actuator power must share a common ground with Arduino.

Inductive loads

Use a transistor/MOSFET driver and flyback diode.

Rewire safely

Disconnect power before moving components or probing continuity.

VISUAL COURSE-PORTAL PATH

Download Course Material

1

SIGN IN

Open lms-tms.tertiaryinfotech.com

2

OPEN COURSE

Select C218

3

DOWNLOAD

Learner Slides, LG and Labs

4

VERIFY

Filename/version match v1.0

01

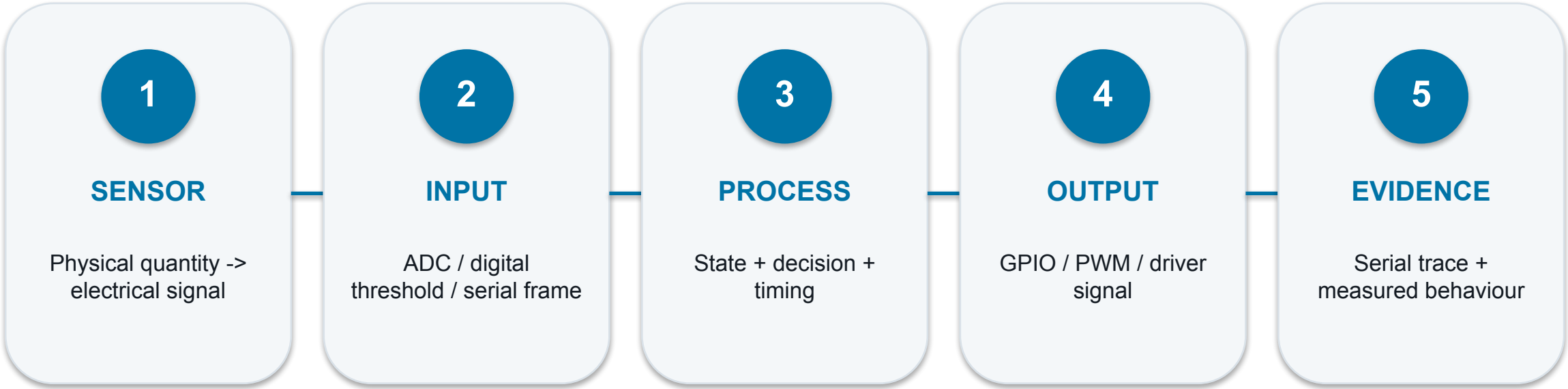
TOPIC 1

Introduction to Embedded Systems

Architecture, resource constraints, ATmega328P, sketch lifecycle, timing, and serial evidence

SIGNAL / CONTROL FLOW

Closed-Loop Embedded System



DECISION / COMPARISON

General-Purpose vs Embedded Design

DESKTOP / MOBILE

Workload

Many user-selected applications

Resources

Large memory/storage; rich OS

Timing

Best effort is often acceptable

Power

Mains or large battery

EMBEDDED CONTROLLER

Workload

Dedicated control function

Resources

Bounded flash, SRAM and I/O

Timing

Deadlines affect physical behaviour

Power

Energy and thermal limits

WORKED CALCULATION

Resource Budget Is a Design Contract

available = capacity - static - worst-case dynamic

SUBSTITUTE REAL VALUES

2,048 B SRAM - 820 B globals - 768 B stack margin = 460 B headroom

Flash

Sketch + libraries <= 32 KB

SRAM

Globals + heap + stack < 2 KB

Timing

Worst-case loop < control deadline

ATmega328P Controller Map

CPU

8-bit AVR core at 16 MHz; executes the control program.

Memory

32 KB flash, 2 KB SRAM, 1 KB EEPROM - each has different persistence.

Interfaces

GPIO, ADC, timers/PWM, UART, SPI and I2C/TWI.

Reset / boot

Bootloader receives sketches; reset restarts setup() then loop().

Arduino Uno Pin Budget

D0/D1

UART RX/TX - avoid conflicts while using Serial.

D3/5/6/9/10/11

PWM-capable on the classic Uno; timer use can interact with libraries.

A0-A5

Six 10-bit ADC channels; A4/A5 also serve I2C.

D13

Built-in LED - useful evidence without external wiring.

SIGNAL / CONTROL FLOW

Sketch Runtime Lifecycle

1

RESET

Registers/peripherals start from reset state

2

BOOTLOADER

Short upload/listen window

3

setup()

One-time pin, serial and library initialisation

4

loop()

Repeated control cycle

5

INTERRUPTS

Asynchronous events alter timing/state

Variables Are Hardware-Relevant

uint8_t

Exact 0-255 range; matches many registers and PWM values.

int

Convenient arithmetic but size is architecture-dependent.

unsigned long

Holds millis()/micros() timestamps without negative values.

const

Documents a fixed pin/threshold and lets the compiler protect intent.

DECISION / COMPARISON

Blocking vs Scheduled Timing

BLOCKING

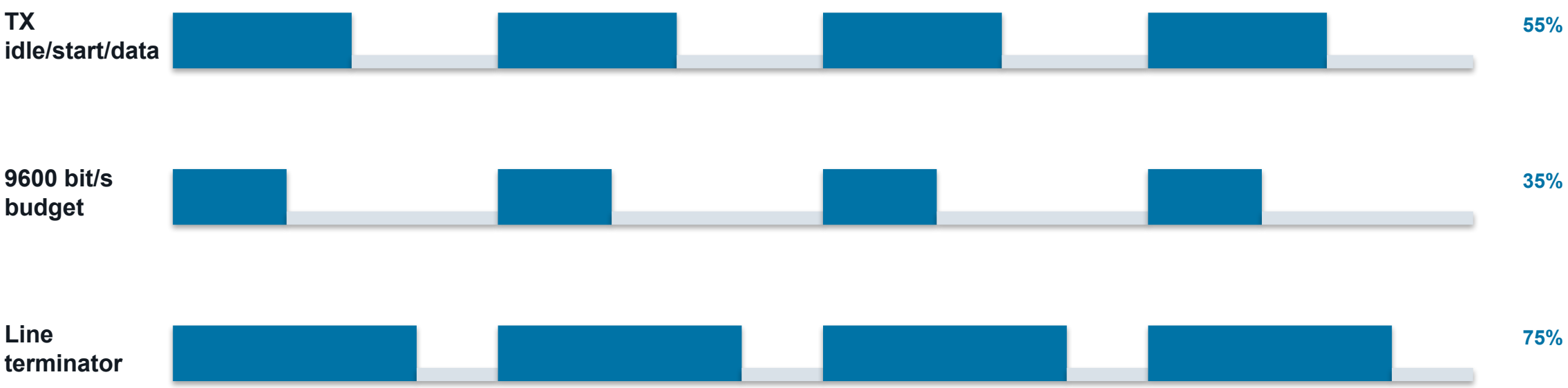
- **delay()**
  Stops normal loop progress
- **Consequence**
  Serial/button work waits
- **Use**
  Simple isolated demonstrations
- **Risk**
  Missed events and slow response

NON-BLOCKING

- **millis() delta**
  Checks elapsed time without stopping
- **Consequence**
  Multiple behaviours interleave
- **Use**
  Responsive control systems
- **Risk**
  Must handle rollover-safe subtraction

TIMING WAVEFORM

UART Frame and Baud Contract



Sender and receiver must agree on baud, framing and line termination; otherwise text may be garbled or commands remain buffered.

Line-Oriented Serial Parser

```
String command;
bool ledOn = false;

void applyOutput() {
  digitalWrite(LED_BUILTIN, ledOn ? HIGH : LOW);
}

void setup() {
  pinMode(LED_BUILTIN, OUTPUT);
  applyOutput();
  Serial.begin(9600);
  Serial.println("READY: ON | OFF | STATUS");
}

void loop() {
  if (Serial.available() == 0) return;
  command = Serial.readStringUntil('\n');
  command.trim();
```

READ THE EVIDENCE

1. readStringUntil('\n') defines one command frame
2. trim() removes CR/LF residue
3. Unknown commands return ERR without changing output
4. Maintained ledOn state makes STATUS truthful

Upload and Runtime Diagnosis

TEST	EXPECTED	OBSERVED	DECISION / ACTION
Compile	Binary produced	Compiler errors	Fix syntax/library
Port open	Board enumerates	No serial port	Cable/driver/permissions
Upload	avrdude success	Timeout / sync error	Board/port/reset
Runtime	Expected LED/Serial trace	Wrong output	State/timing/wiring

Lab 1: Board Check and Deterministic Blink

MECHANISM TO PROVE

Validate the Arduino Uno/Maker UNO toolchain and prove deterministic digital output timing.

DELIVERABLE

A board-identification record and a 250 ms LED heartbeat that verifies compile, upload, reset, and loop execution.

INPUT	CONTROL	OUTPUT	VERIFY
LED_BUILTIN	const uint8_t LED_PIN = LED_	Sketch compiles with no error	Measured ten-cycle duration is approximately 5 seconds

Lab 1 Evidence Code

```
const uint8_t LED_PIN = LED_BUILTIN;
const unsigned long INTERVAL_MS = 250;

void setup() {
  pinMode(LED_PIN, OUTPUT);
}

void loop() {
  digitalWrite(LED_PIN, HIGH);
  delay(INTERVAL_MS);
  digitalWrite(LED_PIN, LOW);
  delay(INTERVAL_MS);
}
```

READ THE EVIDENCE

1. LED_BUILTIN avoids board-specific hard-coding
2. pinMode establishes the output contract
3. HIGH/LOW create two observable states
4. Timing evidence is measured across ten cycles

OBSERVE -> COMPARE -> DECIDE

Lab 1 Acceptance and Failure

TEST	EXPECTED	OBSERVED	DECISION / ACTION
Compile/upload	No errors	Pass/fail log	Fix toolchain before wiring
Period	500 ms per cycle	Ten cycles approx 5 s	Check millisecond units
Reset	Heartbeat restarts	Visible reset sequence	Confirms setup() reruns

Lab 2: Serial Command Control

MECHANISM TO PROVE

Trace UART data from a typed ASCII command to a validated output state.

DELIVERABLE

A line-oriented command parser that accepts ON, OFF, and STATUS without blocking indefinitely.

INPUT	CONTROL	OUTPUT	VERIFY
LED_BUILTIN	String command;	ON lights the LED	Unknown input returns ERR without changing output

EDITABLE STATE MACHINE

Lab 2 Command State



A state machine makes every output, entry condition, exit condition, and safe default inspectable.

Lab 2 Acceptance and Failure

TEST	EXPECTED	OBSERVED	DECISION / ACTION
ON	LED=ON + light	Observed trace	State and output agree
OFF	LED=OFF + dark	Observed trace	State and output agree
BAD	ERR, no change	Observed trace	Parser rejects safely

02

TOPIC 2

Introduction to Basic Electronics

Voltage, current, resistance, breadboards, digital thresholds, pull-ups, debounce, PWM, and protected loads

WORKED CALCULATION

LED Current-Limit Resistor

R = (Vsource - Vf) / I

SUBSTITUTE REAL VALUES

R = (5.0 V - 2.0 V) / 0.010 A = 300 ohm -> choose 330 ohm

Voltage

Resistor drops about 3 V

Current

About 9 mA with 330 ohm

Power

P = I^2R approx 0.027 W

DECISION / COMPARISON

Series vs Parallel Circuits

SERIES

- **Current**
  Same current through each element
- **Voltage**
  Drops add to source voltage
- **Open fault**
  Breaks the whole path
- **Example**
  LED + current-limit resistor

PARALLEL

- **Voltage**
  Same voltage across branches
- **Current**
  Branch currents add
- **Open fault**
  Other branches may continue
- **Example**
  Multiple independently driven loads

Breadboard Connectivity

Terminal strips

Groups of five holes are internally common; the centre gap isolates sides.

Power rails

Long rails distribute supply but may be split mid-board.

Cross-check

Use continuity mode before power when rail segmentation is uncertain.

Failure signature

Same-row component legs can short a device or bypass a resistor.

Polarity and Component Boundaries

LED

Anode to controlled positive path; cathode to return.

Diode

Cathode band identifies direction; flyback diode is reverse-biased normally.

Electrolytic capacitor

Observe + and - markings; reverse polarity can damage the part.

Transistor/MOSFET

Pin order varies by device; verify the datasheet, not package appearance.

Digital Input Thresholds

LOW

Voltage below the input-low threshold is read as logic 0.

HIGH

Voltage above the input-high threshold is read as logic 1.

Undefined region

Intermediate voltage is not a valid logic contract.

Floating

An unconnected high-impedance input can pick up noise and change randomly.

SIGNAL / CONTROL FLOW

INPUT_PULLUP Signal Path

1

INTERNAL RESISTOR

Weak pull-up connects input to VCC

2

RELEASED

Input reads HIGH

3

PRESSED

Switch connects input to GND

4

SOFTWARE

Interpret LOW as active

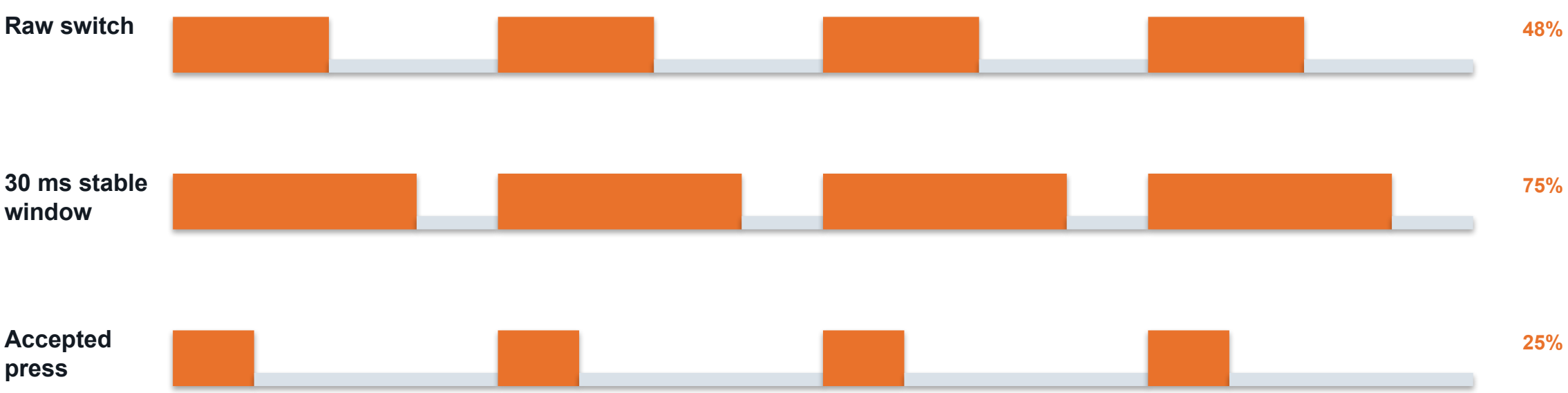
5

EVIDENCE

Raw + debounced states

TIMING WAVEFORM

Mechanical Bounce vs Debounced State



The controller accepts a new state only after the raw input remains unchanged for the debounce interval.

Edge Detection and Debounce

```
const uint8_t BUTTON_PIN = 2;
const uint8_t LED_PIN = 7;
const unsigned long DEBOUNCE_MS = 30;
bool ledOn = false;
bool stableState = HIGH;
bool lastRaw = HIGH;
unsigned long changedAt = 0;

void setup() {
  pinMode(BUTTON_PIN, INPUT_PULLUP);
  pinMode(LED_PIN, OUTPUT);
}

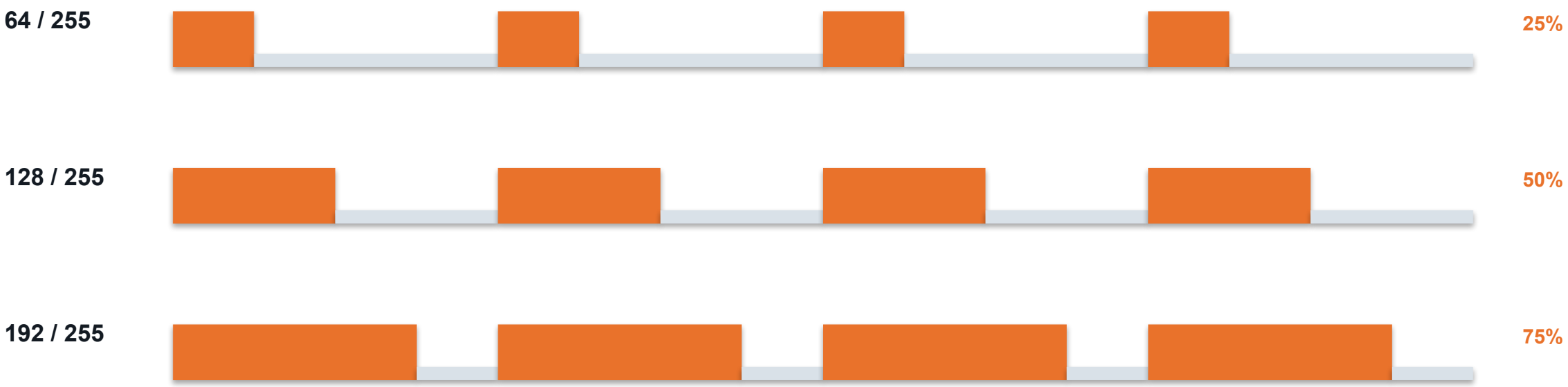
void loop() {
  bool raw = digitalRead(BUTTON_PIN);
  if (raw != lastRaw) { changedAt = millis(); lastRaw = raw; }
  if (millis() - changedAt >= DEBOUNCE_MS && raw != stableState) {
    stableState = raw;
```

READ THE EVIDENCE

1. lastRaw detects any electrical transition
2. changedAt records when stability began
3. stableState changes only after 30 ms
4. Only the confirmed falling edge toggles the LED

TIMING WAVEFORM

PWM Duty Cycle



On the classic Uno, analogWrite() produces PWM on supported pins; it is not a true analogue voltage.

Uno PWM and Timer Boundaries

8-bit command

analogWrite value 0-255 maps to 0-100% duty cycle.

PWM pins

Classic Uno: D3, D5, D6, D9, D10, D11.

Timer sharing

Tone, Servo, and timing libraries can change timer availability/behaviour.

Load boundary

PWM pin drives a control input; a transistor stage drives higher current.

SIGNAL / CONTROL FLOW

Protected Inductive Load

1

GPIO/PWM

Low-current logic signal

2

GATE/BASE

Current-limited drive

3

MOSFET/ TRANSISTOR

Switches motor current

4

FLYBACK DIODE

Clamps inductive voltage

5

SUPPLY

Separate load energy + common ground

Lab 3: Safe LED Circuit and Ohm's Law

MECHANISM TO PROVE

Calculate and build a current-limited LED circuit within an Arduino digital pin's electrical boundary.

DELIVERABLE

A documented 5 V digital-output circuit using a 330 ohm resistor and measured LED voltage.

INPUT	CONTROL	OUTPUT	VERIFY
D7	const uint8_t LED_PIN = 7;	LED polarity is correct	Calculated current remains close to or below 10 mA

OBSERVE -> COMPARE -> DECIDE

Lab 3 Electrical Evidence

TEST	EXPECTED	OBSERVED	DECISION / ACTION
R calc	300 ohm minimum target	330 ohm fitted	Safe standard value
LED Vf	Approx 1.8-2.2 V	Multimeter value	Use resistor voltage for current
I/O state	D7 HIGH/LOW	LED ON/OFF	Pin contract proven

Lab 4: Push Button with INPUT_PULLUP

MECHANISM TO PROVE

Eliminate a floating input and debounce a mechanical switch using a stable time window.

DELIVERABLE

An active-LOW button that toggles an LED once per physical press without double triggers.

INPUT	CONTROL	OUTPUT	VERIFY
D2	const uint8_t BUTTON_PIN = 2	Released input reads HIGH	Bounce within 30 ms does not create extra toggles

Lab 4 Debounced Button



A state machine makes every output, entry condition, exit condition, and safe default inspectable.

Lab 5: PWM Fade with Non-blocking Timing

MECHANISM TO PROVE

Relate 8-bit PWM values to duty cycle and vary LED brightness without blocking other logic.

DELIVERABLE

A triangular 0-255 PWM ramp on D9 with calculated duty-cycle checkpoints.

INPUT	CONTROL	OUTPUT	VERIFY
D9 (~)	const uint8_t LED_PIN = 9;	LED fades up and down continuously	Calculated 128 duty cycle is approximately 50%

Lab 5 Non-blocking PWM

```
const uint8_t LED_PIN = 9;
const unsigned long UPDATE_MS = 20;
const int STEP = 5;
int brightness = 0;
int direction = STEP;
unsigned long lastUpdate = 0;

void setup() { pinMode(LED_PIN, OUTPUT); }

void loop() {
  if (millis() - lastUpdate < UPDATE_MS) return;
  lastUpdate = millis();
  brightness += direction;
  if (brightness >= 255) { brightness = 255; direction = -STEP; }
  if (brightness <= 0) { brightness = 0; direction = STEP; }
  analogWrite(LED_PIN, brightness);
}
```

READ THE EVIDENCE

1. Elapsed-time test keeps loop responsive
2. direction carries the ramp sign
3. Boundary clamp prevents wraparound
4. analogWrite commits the 8-bit duty value

OBSERVE -> COMPARE -> DECIDE

Topic 2 Verification Matrix

TEST	EXPECTED	OBSERVED	DECISION / ACTION
Digital output	Safe LED current	Measured resistor/Vf	Safe interconnection
Digital input	Stable active-LOW	Raw vs debounced trace	Signal conditioning
PWM	0-255 bounded ramp	Duty checkpoints	PWM mechanism

03

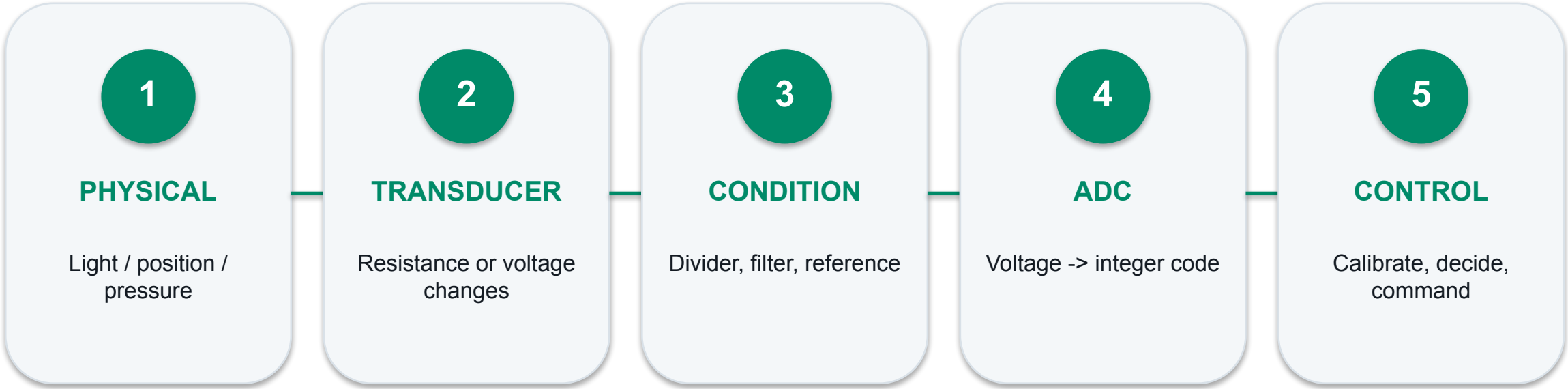
TOPIC 3

Analogue Sensors and Transducers

Signal chains, 10-bit ADC, quantisation, calibration, filtering, LDR dividers, hysteresis, and servo control

SIGNAL / CONTROL FLOW

Analogue Measurement Chain



Continuous Signal, Discrete Code

Input

A0 accepts a voltage within the board's permitted range.

Sample

ADC observes the input at a point in time.

Quantise

The continuous value is assigned to one of 1024 codes.

Report

analogRead() returns an integer from 0 to 1023 on the classic Uno.

WORKED CALCULATION

10-bit ADC Conversion

Vin approx code x Vref / 1023

SUBSTITUTE REAL VALUES

code 614 at Vref 5.00 V -> Vin approx 3.00 V

Resolution

5 V / 1024 approx 4.88 mV per count

Range

Never exceed the allowed analogue input voltage

Reference

Measurement accuracy depends on actual Vref

Reference Voltage and Accuracy

Default reference

Typically the board supply; USB/supply variation affects conversion.

Source impedance

High impedance can slow sample/hold settling.

Noise

Long wires and motors inject variation into ADC readings.

Calibration

Use measured endpoints rather than assuming ideal 0 and 1023.

Moving-Average Filter

```
const int N = 8; long sum = 0;
for (int i=0; i<N; i++) {
  sum += analogRead(A0);
}
int filtered = sum / N;
```

READ THE EVIDENCE

1. Eight independent samples contribute
2. long avoids sum overflow in larger windows
3. Division produces the mean code
4. Filtering reduces noise but adds latency

WORKED CALCULATION

Calibrate and Map a Sensor

output = map(raw, rawMin, rawMax, outMin, outMax)

SUBSTITUTE REAL VALUES

map(constrain(raw, 180, 850), 180, 850, 0, 100)

Measure

Capture real minimum/maximum

Constrain

Reject out-of-range excursions

Validate

Compare mapped value to a reference

WORKED CALCULATION

LDR Voltage Divider

Vout = Vcc x Rfixed / (RLDR + Rfixed)

SUBSTITUTE REAL VALUES

5 V x 10k / (30k + 10k) = 1.25 V in a darker condition

Orientation

Swapping LDR/fixed resistor reverses response

ADC

A0 observes divider junction

Threshold

Calibrate with measured light levels

EDITABLE STATE MACHINE

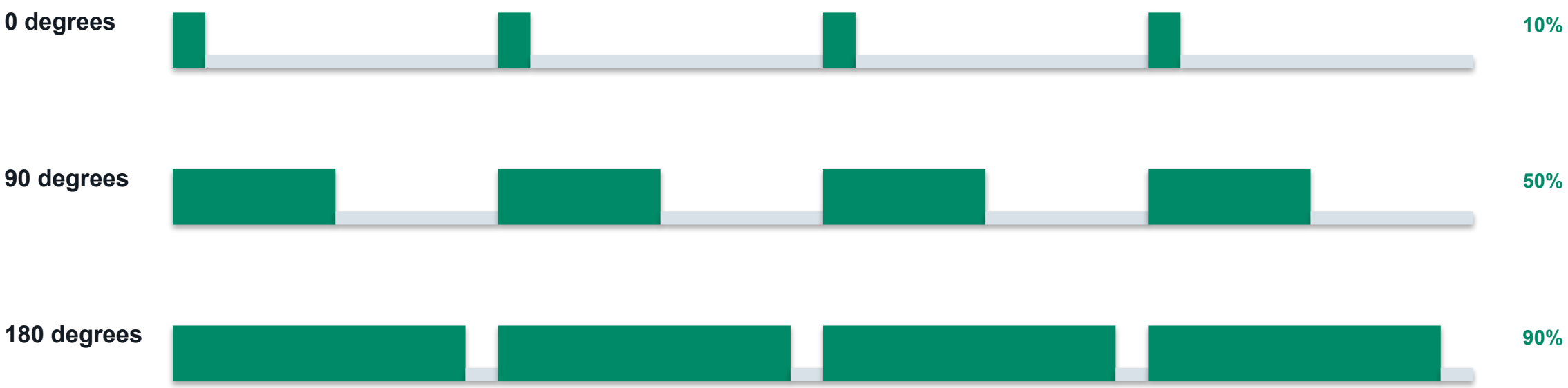
Hysteresis Prevents Chatter



A state machine makes every output, entry condition, exit condition, and safe default inspectable.

TIMING WAVEFORM

Servo Command Pulse



The Servo library schedules control pulses; servo supply current must not come from an Arduino I/O pin.

SIGNAL / CONTROL FLOW

Servo Energy and Signal Boundary

1

A0

Potentiometer position

2

map()

0-1023 -> 0-180

3

D9 signal

Timed control pulse

4

5 V supply

Provides actuator energy

5

COMMON GND

Maintains signal reference

OBSERVE -> COMPARE -> DECIDE

Analogue Failure Modes

TEST	EXPECTED	OBSERVED	DECISION / ACTION
Stuck 0	Input shorted to GND	ADC trace	Inspect divider
Stuck 1023	Input tied to Vref	ADC trace	Inspect return path
Jitter	Noise / weak supply	Spread over samples	Filter / power
Wrong direction	Divider/map reversed	Endpoint test	Reverse mapping

Lab 6: LDR Dark-Activated Lamp with Hysteresis

MECHANISM TO PROVE

Convert a light-dependent voltage divider into a stable two-threshold control decision.

DELIVERABLE

A calibrated dark-activated lamp that avoids chatter around the switching threshold.

INPUT	CONTROL	OUTPUT	VERIFY
5 V	const uint8_t LDR_PIN = A0;	Serial output shows plausible 0-1023 readings	Output remains stable inside the hysteresis band

OBSERVE -> COMPARE -> DECIDE

Lab 6 Calibration Evidence

TEST	EXPECTED	OBSERVED	DECISION / ACTION
Bright	Record ADC high/low	Measured value	Choose orientation
Room	Record midpoint	Measured value	Set dead band
Covered	Record dark endpoint	Measured value	Set ON threshold

Lab 7: Potentiometer-Controlled Servo

MECHANISM TO PROVE

Map a 10-bit ADC measurement to a bounded servo command while respecting the servo power boundary.

DELIVERABLE

A 0-180 degree servo position controller with clamped input and separate-power guidance.

INPUT	CONTROL	OUTPUT	VERIFY
Pot outer pins	#include <Servo.h>	ADC reading changes smoothly	Board does not reset when servo moves

Lab 7 ADC-to-Servo Contract

```
#include <Servo.h>
Servo actuator;
const uint8_t POT_PIN = A0;
const uint8_t SERVO_PIN = 9;

void setup() {
  actuator.attach(SERVO_PIN);
  Serial.begin(9600);
}

void loop() {
  int raw = analogRead(POT_PIN);
  int angle = constrain(map(raw, 0, 1023, 0, 180), 0, 180);
  actuator.write(angle);
  Serial.println(angle);
  delay(15);
}
```

READ THE EVIDENCE

1. analogRead acquires the 10-bit measurement
2. map converts measurement to command units
3. constrain enforces actuator command bounds
4. Common ground makes the signal meaningful

SIGNAL / CONTROL FLOW

Analogue Control Case

1

MEASURE

Potentiometer / LDR

2

CALIBRATE

Real endpoints

3

FILTER

Reduce noise

4

DECIDE

Map or hysteresis

5

ACT

Servo / lamp

OBSERVE -> COMPARE -> DECIDE

Topic 3 Verification Matrix

TEST	EXPECTED	OBSERVED	DECISION / ACTION
ADC scale	0-1023 bounded	Serial endpoints	ADC mechanism
LDR control	Stable dark lamp	Threshold trace	Applied sensor
Servo	0-180 bounded	Angle checkpoints	Transducer control

04

TOPIC 4

Digital Sensors and Transducers

Digital sensor contracts, trigger/echo timing, distance calculation, timeouts, PIR state, and responsive control

Digital Sensor Output Contracts

Binary level

Sensor presents HIGH/LOW relative to logic thresholds.

Pulse width

Information is encoded in how long a signal remains HIGH/LOW.

Serial protocol

Bits are framed with timing, addressing and sometimes checksums.

Library object

A library can abstract timing but errors still require explicit handling.

SIGNAL / CONTROL FLOW

Ultrasonic Ranging Transaction

1

IDLE LOW

Settle trigger

2

10 us TRIGGER

Start burst

3

40 kHz BURST

Sensor transmits

4

ECHO HIGH

Pulse equals round trip

5

TIMEOUT

Bound missing response

WORKED CALCULATION

Time-of-Flight Distance

distance = duration x 0.0343 cm/us / 2

SUBSTITUTE REAL VALUES

1,750 us x 0.0343 / 2 = 30.0 cm

Round trip

Divide by two

Temperature

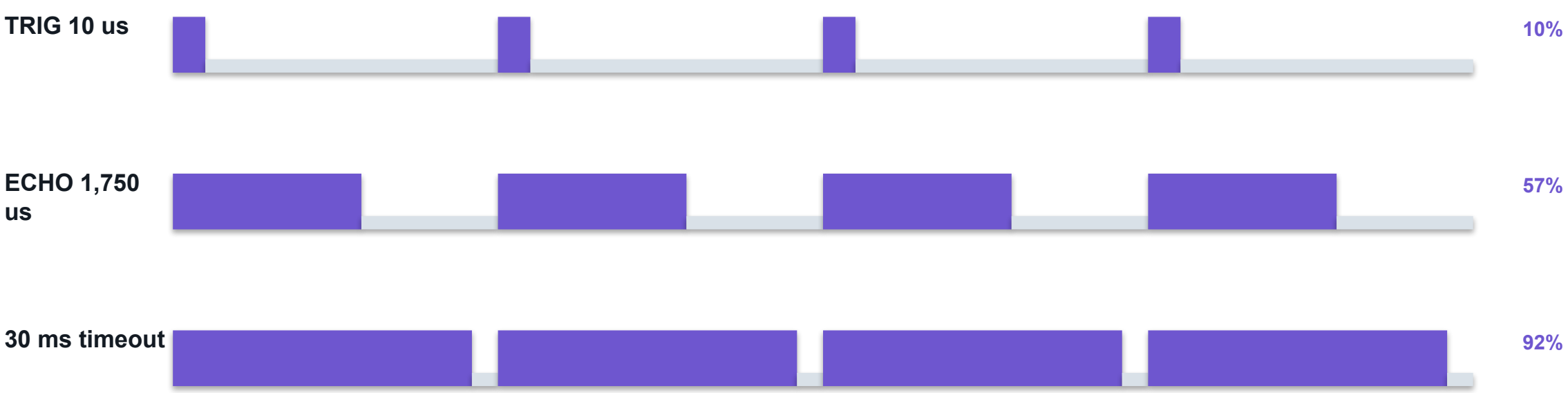
Sound speed changes slightly

Timeout

0 means no valid echo

TIMING WAVEFORM

Trigger and Echo Timing



pulseIn(ECHO, HIGH, 30000) converts a missing echo into duration == 0 instead of an unbounded wait.

Bounded pulseIn() Measurement

```
const uint8_t TRIG_PIN = 9;
const uint8_t ECHO_PIN = 10;
const unsigned long ECHO_TIMEOUT_US = 30000;

void setup() {
  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);
  Serial.begin(9600);
}

void loop() {
  digitalWrite(TRIG_PIN, LOW); delayMicroseconds(2);
  digitalWrite(TRIG_PIN, HIGH); delayMicroseconds(10);
  digitalWrite(TRIG_PIN, LOW);
  unsigned long duration = pulseIn(ECHO_PIN, HIGH, ECHO_TIMEOUT_US);
  if (duration == 0) Serial.println("NO_ECHO");
  else {
    float distanceCm = duration * 0.0343f / 2.0f;
```

READ THE EVIDENCE

1. 2 us LOW establishes a clean trigger edge
2. 10 us HIGH starts the ranging cycle
3. 30,000 us timeout bounds waiting
4. duration == 0 is a separate fault state

Ultrasonic Measurement Limits

Blind zone

Very near targets may not produce a usable separated echo.

Surface angle

A slanted target reflects energy away from the receiver.

Material

Soft or irregular surfaces absorb/scatter ultrasound.

Cross-talk

Nearby ultrasonic sensors can hear another transmitter.

SIGNAL / CONTROL FLOW

PIR Motion Contract

1

WARM-UP

Sensor stabilises

2

IDLE

OUT LOW

3

MOTION

OUT HIGH

4

HOLD TIMER

Extend on new motion

5

EXPIRE

Return light LOW

Digital Sensor Noise Controls

Pull-up/down

Defines idle voltage when output is open or high impedance.

Debounce

Rejects rapid contact transitions.

Timeout

Bounds missing pulses or stalled devices.

State machine

Separates warm-up, active, hold and fault behaviour.

DECISION / COMPARISON

delay() Hold vs millis() Hold

BLOCKING HOLD

Control

delay(5000)

Effect

No normal sensor/serial work

Retrigger

Cannot extend cleanly

Use

Simple one-shot demo

RESPONSIVE HOLD

Control

elapsed < HOLD_MS

Effect

Loop remains responsive

Retrigger

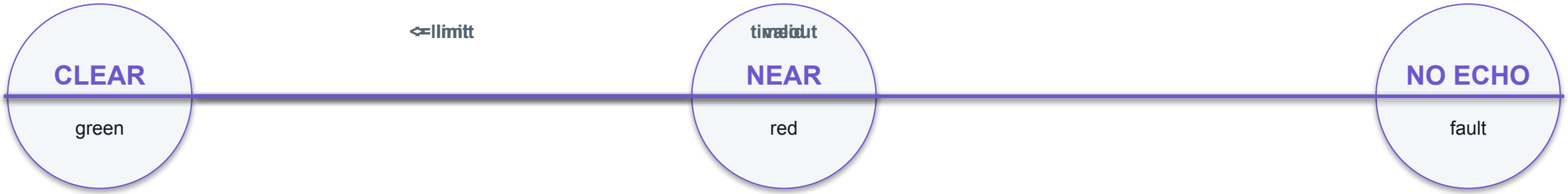
Update lastMotionMs

Use

Real motion light

EDITABLE STATE MACHINE

Obstacle Indicator States



A state machine makes every output, entry condition, exit condition, and safe default inspectable.

Lab 8: Ultrasonic Distance with Timeout

MECHANISM TO PROVE

Measure echo pulse width, convert time-of-flight to distance, and handle a missing echo safely.

DELIVERABLE

A distance monitor that reports centimetres or NO_ECHO within a bounded 30 ms wait.

INPUT	CONTROL	OUTPUT	VERIFY
Sensor VCC	const uint8_t TRIG_PIN = 9;	Distance is within a reasonable tolerance of the ruler	No divide-by-zero or indefinite wait occurs

Lab 8 Distance Evidence

```
const uint8_t TRIG_PIN = 9;
const uint8_t ECHO_PIN = 10;
const unsigned long ECHO_TIMEOUT_US = 30000;

void setup() {
  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);
  Serial.begin(9600);
}

void loop() {
  digitalWrite(TRIG_PIN, LOW); delayMicroseconds(2);
  digitalWrite(TRIG_PIN, HIGH); delayMicroseconds(10);
  digitalWrite(TRIG_PIN, LOW);
  unsigned long duration = pulseIn(ECHO_PIN, HIGH, ECHO_TIMEOUT_US);
  if (duration == 0) Serial.println("NO_ECHO");
  else {
    float distanceCm = duration * 0.0343f / 2.0f;
```

READ THE EVIDENCE

1. TRIG and ECHO have distinct directions
2. Echo time is unsigned long microseconds
3. Formula divides round-trip path by two
4. NO_ECHO is not treated as zero distance

OBSERVE -> COMPARE -> DECIDE

Lab 8 Acceptance Evidence

TEST	EXPECTED	OBSERVED	DECISION / ACTION
20 cm	Approx 20 cm	Serial + ruler	Within tolerance
80 cm	Approx 80 cm	Serial + ruler	Within tolerance
No target	NO_ECHO	Serial state	Timeout handled

Lab 9: PIR Motion-Activated Light

MECHANISM TO PROVE

Use a digital motion sensor contract and a non-blocking hold timer to control a light.

DELIVERABLE

A motion light that extends its ON period when new motion arrives and expires safely after five seconds.

INPUT	CONTROL	OUTPUT	VERIFY
PIR VCC	const uint8_t PIR_PIN = 2;	Motion drives the output HIGH	loop() remains responsive without delay(5000)

EDITABLE STATE MACHINE

Lab 9 Motion-Light Control



A state machine makes every output, entry condition, exit condition, and safe default inspectable.

OBSERVE -> COMPARE -> DECIDE

Lab 9 Acceptance Evidence

TEST	EXPECTED	OBSERVED	DECISION / ACTION
Warm-up	Stable idle LOW	Observed OUT	Ready
Motion	Light HIGH	LED evidence	Sensor detected
Retrigger	Hold extends	Timestamp trace	Non-blocking

OBSERVE -> COMPARE -> DECIDE

Digital Sensor Fault Isolation

TEST	EXPECTED	OBSERVED	DECISION / ACTION
Power	Correct VCC/GND	Meter / module LED	Fix supply
Signal	Expected HIGH/LOW/pulse	Serial/raw trace	Fix pin/threshold
Timing	Within protocol window	Logic/waveform	Fix timeout/edge
Control	State transition correct	Output evidence	Fix decision logic

Digital Sensor Decision Rules

Know the encoding

Level, pulse width, serial frame, or library abstraction.

Bound every wait

A failed sensor must not freeze the control loop.

Represent faults

NO_ECHO/READ_ERROR are states, not plausible measurements.

Verify with references

Compare sensor evidence against a ruler, known motion, or protocol timing.

05

TOPIC 5

Actuators

Actuator energy boundaries, motor drivers, flyback protection, steppers, DHT validation, and feedback control

SIGNAL / CONTROL FLOW

Actuator Control Architecture



TECHNICAL MODEL

Why Motors Need a Driver

Current

Motor startup/stall current exceeds an I/O pin's safe capability.

Inductance

Switch-off generates a voltage spike that can damage electronics.

Noise

Brushes and current steps disturb supply and sensor signals.

Isolation of roles

GPIO commands; the driver and external supply deliver energy.

SIGNAL / CONTROL FLOW

Protected DC Motor Path

1

D5 PWM

Control duty

2

GATE RESISTOR

Limit edge current

3

MOSFET

Low-side switch

4

FLYBACK DIODE

Clamp inductive kick

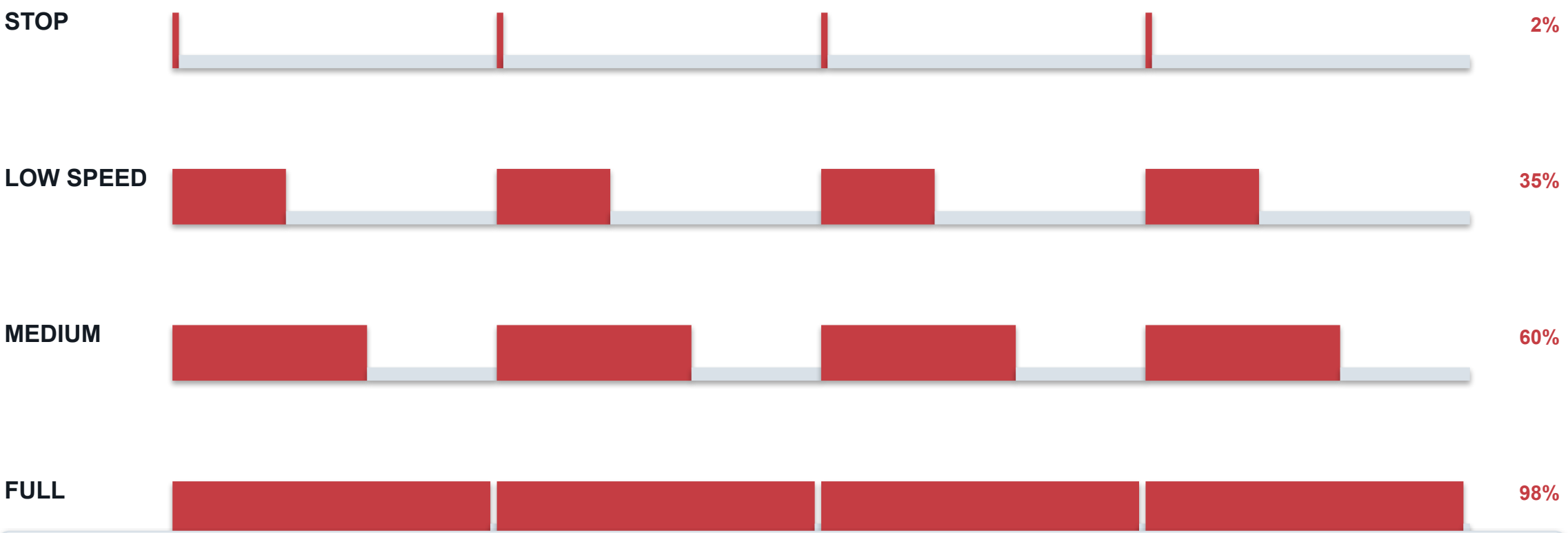
5

MOTOR SUPPLY

Deliver load current

TIMING WAVEFORM

PWM Motor Command



Commanded duty controls average drive, but actual speed depends on load, supply, motor characteristics and the driver's losses.

Stepper Position Mechanism

Phase sequence

Coils energise in a defined order to create incremental motion.

Step count

Position is inferred from commanded steps when no steps are missed.

Speed/torque tradeoff

Excessive step rate causes vibration or lost steps.

Driver array

ULN2003 channels sink coil current and include protection diodes.

SIGNAL / CONTROL FLOW

ULN2003 Stepper Boundary

1

D8-D11

Four logic phases

2

IN1-IN4

Driver inputs

3

DARLINGTON

Current amplification

4

COM DIODES

Inductive protection

5

28BYJ-48

Geared position output

DHT11 Transaction and Validity

Command/read

The library manages the sensor's timing-sensitive exchange.

Sample interval

The device is slow; repeated reads must respect its timing.

NaN

A failed transaction is not a temperature or humidity value.

Control gate

Actuator decisions run only when the measurement is valid.

Reject Invalid Measurements

```
float t = dht.readTemperature();
float h = dht.readHumidity();
if (isnan(t) || isnan(h)) {
  Serial.println("READ_ERROR");
  return;
}
// Only valid values reach control logic.
```

READ THE EVIDENCE

1. Acquire both measurements
2. isnan() detects failed conversion
3. Fault state is reported explicitly
4. return prevents unsafe false control

EDITABLE STATE MACHINE

Fan Hysteresis Controller



A state machine makes every output, entry condition, exit condition, and safe default inspectable.

Actuator Failure Modes

TEST	EXPECTED	OBSERVED	DECISION / ACTION
No motion	No supply/drive	Measure supply/gate	Fix power stage
Resets	Supply dip/noise	Serial reset trace	Separate/decouple
Hot driver	Wrong device/current	Temperature/current	Reduce load
Lost steps	Rate too high	Position error	Lower speed

Lab 10: DHT11 Temperature and Humidity Monitor

MECHANISM TO PROVE

Read a library-managed digital sensor transaction and reject invalid measurements before control use.

DELIVERABLE

A one-hertz temperature/humidity monitor that identifies NaN failures instead of treating them as real values.

INPUT	CONTROL	OUTPUT	VERIFY
DHT VCC	#include <DHT.h>	Temperature and humidity update approximately once per second	dht.begin() is called in setup()

OBSERVE -> COMPARE -> DECIDE

Lab 10 DHT Evidence

TEST	EXPECTED	OBSERVED	DECISION / ACTION
Room	Plausible T/RH	Serial trace	Valid
Warm	Temperature rises	Serial trace	Sensor responds
Data open	READ_ERROR	Fault trace	Invalid rejected

Lab 11: Stepper Motor Position Control

MECHANISM TO PROVE

Drive a four-phase stepper through a transistor array and relate commanded steps to angular position.

DELIVERABLE

A 28BYJ-48/ULN2003 actuator that moves one revolution forward and backward at a bounded speed.

INPUT	CONTROL	OUTPUT	VERIFY
D8-D11	#include <Stepper.h>	Motor completes a forward and reverse move	Grounds are common and board remains stable

SIGNAL / CONTROL FLOW

Lab 11 Phase-to-Position Chain

1

STEPS

+2048 / -2048

2

LIBRARY

Phase order

3

ULN2003

Coil current

4

GEARBOX

Mechanical reduction

5

EVIDENCE

Marked shaft position

Lab 11 Acceptance Evidence

TEST	EXPECTED	OBSERVED	DECISION / ACTION
Forward	One revolution	Shaft mark	Position close
Reverse	Return to origin	Shaft mark	Direction proven
Higher speed	Possible missed steps	Position error	Set safe rate

Lab 12: DHT Fan Controller with Hysteresis

MECHANISM TO PROVE

Integrate measurement, decision logic, and a protected continuous-drive actuator into a feedback controller.

DELIVERABLE

A temperature-controlled DC fan with MOSFET switching, flyback protection, PWM drive, and hysteresis.

INPUT	CONTROL	OUTPUT	VERIFY
DHT DATA	#include <DHT.h>	Fan turns ON above 30 C	Motor current never flows through an Arduino I/O pin

Lab 12 Feedback Controller

```
#include <DHT.h>
const uint8_t DHT_PIN = 2;
const uint8_t FAN_PWM_PIN = 5;
const float FAN_ON_C = 30.0;
const float FAN_OFF_C = 28.0;
DHT dht(DHT_PIN, DHT11);
bool fanOn = false;

void setup() {
  pinMode(FAN_PWM_PIN, OUTPUT);
  Serial.begin(9600);
  dht.begin();
}

void loop() {
  float t = dht.readTemperature();
  if (isnan(t)) { Serial.println("READ_ERROR"); delay(2000); return;
}
  if (!fanOn && t >= FAN_ON_C) fanOn = true;
```

READ THE EVIDENCE

1. Invalid sensor value exits before control
2. Separate ON/OFF thresholds prevent chatter
3. PWM pin commands the MOSFET, not the motor directly
4. Serial trace records measurement and actuator state

OBSERVE -> COMPARE -> DECIDE

Lab 12 Acceptance Evidence

TEST	EXPECTED	OBSERVED	DECISION / ACTION
>=30 C	Fan ON	T/FAN trace	Control enters ON
28-30 C	Hold prior state	T/FAN trace	Hysteresis
<=28 C	Fan OFF	T/FAN trace	Control exits ON

Course Verification Matrix

TEST	EXPECTED	OBSERVED	DECISION / ACTION
LO1	Board/serial mechanism	Labs 1-2	Topic 1
LO2	Safe circuits/input/PWM	Labs 3-5	Topic 2
LO3	ADC/LDR/servo	Labs 6-7	Topic 3
LO4	Ultrasonic/PIR	Labs 8-9	Topic 4

CLOSE-OUT

Course Feedback

Course evaluation

Complete the feedback form in LMS/TMS.

Your comments

Tell the trainer what worked and what to improve.

Completion

Confirm the portal records your feedback.

Support

Keep the Learner Guide and labs for post-course practice.

AFTER-CLASS HELP

Support

Email

enquiry@tertiaryinfotech.com

Telephone

+65 6100 0613

Course portal

https://lms-tms.tertiaryinfotech.com/

Arduino reference

https://docs.arduino.cc/language-reference/

SENSE | DECIDE | ACT | VERIFY

Thank You

A working prototype is a claim.
A measured, repeatable acceptance test is evidence.